LAB 10 - PROMPTS

Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab10-strategy-grid-search/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: Define the search space (A2, K2)

Do this (no prompt needed)
Read GRIDS in Lab10-strategy-grid-search/solution/search.py: sma_cross fast x slow (24 valid pairs), rsi_reversion n x entry x exit (36), bollinger_breakout window x k (16) - 76 sets.

Check: You can say why fast 50 / slow 50 is skipped and why every extra set increases the chance of a lucky winner.

Step 2: Build the grid search (A2, A4)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/search.py. IS_END = '2021-12-31', OOS_START = '2022-01-01'. GRIDS = {'sma_cross': {'fast': [5,10,20,30,50], 'slow': [50,100,150,200,250]}, 'rsi_reversion': {'n': [2,3,5,14], 'entry': [10,20,30], 'exit': [50,60,70]}, 'bollinger_breakout': {'window': [10,20,30,50], 'k': [1.0,1.5,2.0,2.5]}}. grid_points(name) yields parameter dicts (skip sma_cross with fast >= slow). search(bars, costs) runs ONE backtest per set with pipeline.signals.target_position and common.backtest.backtest (signals are causal, so one run covers both windows), then records IS and OOS Sharpe and CAGR (common.metrics) and trades; add IS_rank and OOS_rank (method='min', NaN last) and decay = OOS - IS Sharpe. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: search(...) returns a DataFrame with 76 rows.

Step 3: Add ranking, robustness and the decay report (A4, K2)

Prompt - copy into your AI coding assistant
In labs_fullcode/pipeline/search.py add robust_choice(ma) - for the sma_cross grid, the cell with the best AVERAGE in-sample Sharpe over itself and its neighbours - and run(symbol='SPY') for `python -m pipeline search`: print buy-and-hold Sharpe IS and OOS; a decay report with the best IS set per family (IS -> OOS Sharpe, decay, OOS rank of 76, trades); the top 5 by IS Sharpe; the Spearman rank correlation between IS and OOS Sharpe; the average decay of the top 5; the share of all sets that beat buy and hold OOS; the single best MA cell versus the robust MA region; and a suggested config.yaml strategy line (do NOT edit the config automatically). Save _outputs/Lab10_grid_results.csv, _outputs/Lab10_decay_report.csv and chart Lab10_grid_heatmap.png (IS and OOS Sharpe heatmaps of the MA grid with values in each cell). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline search --help shows --symbol.

Step 4: Run the search (A4)

Do this (no prompt needed)
Run: python -m pipeline search --symbol SPY (same as: python3 Lab10-strategy-grid-search/search_stage.py).

Check: 76 sets; buy-and-hold Sharpe 0.87 IS and 0.67 OOS. Best per family: sma_cross 10/100 IS 1.23 -> OOS 0.70 (decay -0.53, OOS rank 26); rsi_reversion n 2, entry 30, exit 70: 0.60 -> 0.74; bollinger_breakout 20 / 1.0: 0.91 -> 1.04 (OOS rank 3).

Step 5: Read the overfitting evidence (A4, K2)

Do this (no prompt needed)
Read the top-5 table and the three summary lines.

Check: Top 5 are all sma_cross; average decay -0.62 Sharpe points; IS/OOS rank correlation 0.06 (the in-sample ranking has almost no predictive value); 43% of all sets beat buy and hold out of sample.

Step 6: Prefer a robust region to a single peak (A2, A4)

Do this (no prompt needed)
Compare the single best MA cell with the robust MA region and open _outputs/Lab10_grid_heatmap.png.

Check: Best cell 10/100: IS 1.23 -> OOS 0.70; robust region 5/50: neighbour-average IS 1.04, own IS 0.92 -> OOS 0.44 - robustness reduces luck but guarantees nothing.

Step 7: Decide whether to promote anything (A2, K2)

Prompt - copy into your AI coding assistant
Using labs_fullcode/_outputs/Lab10_decay_report.csv and labs_fullcode/_outputs/Lab10_grid_results.csv, argue for and against replacing the pipeline's configured ema_cross 10/30 with the best sma_cross or bollinger_breakout set. Mention the number of sets tried (76), the decay, the rank correlation and what the diagnostics stage (Lab 18) should check before any change. Finish with a one-line recommendation.

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Check: A balanced recommendation that keeps ema_cross until a diagnostics run and an A/B test (Lab 13) support a change.

Run it

Step	Command / file
Run Lab 10	python3 Lab10-strategy-grid-search/search_stage.py (same as: python -m pipeline search)
The complete code	pipeline/search.py (reference copy: Lab10-strategy-grid-search/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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